Prospect choice is generally viewed as a game against nature. This article models prospect choice as an n-*person game where each subject assumes that* n-*1 others will be exposed to the same decision problem (prospect choice set) as self; the goal is not to "beat nature" but to do* relatively better *than rivals exposed to the same problem. Preference becomes strategy choice in* n-*person Nash equilibrium. When symmetric pure strategy equilibria do not exist, choice is a symmetric randomized equilibrium; here, uncertainty (probabilistic response) becomes a method of dealing with uncertainty in nature. The approach produces, qualitatively, several empirical expected utility paradoxes (the certainty effect, intransitive cycles, and one form of reflection), and an evolutionary game-theoretic extension accounts for all the phenomena revealed by the research of Kahneman and Tversky.*

Gambling in a Malthusian Universe

A GAME-THEORETIC APPROACH TO THE PARADOXES OF EXPECTED UTILITY

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1. INTRODUCTION: THE PARADOXES OF EXPECTED UTILITY

Decision theory rests on the assumption that, at any given moment, people exhibit preferences that are complete and transitive (Luce and Raiffa 1957; Savage 1954; von Neumann and Morgenstern 1953). That is, for all prospects

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A and B, either $A \geq B$ or $B \geq A$, where "$\geq$" denotes "is preferred or indifferent to" ("$>$" will denote strict preference); and, given any three prospects A, B, C, if $A \geq B$ and $B \geq C$, then $A \geq C$. In economics, preferences are usually represented as a real valued utility function over a set of prospects that is "linear in the probabilities" (Luce and Raiffa 1957; Machina 1982, 1987; Savage 1954; Schoemaker 1982; von Neumann and Morgenstern 1953). Given a discrete probability distribution $(p_1, p_2, \ldots, p_N)$ associated with outcomes $(x_1, x_2, \ldots, x_N)$, the expected utility of the distribution is simply $\Sigma\, u(x_i)p_i$, where $u(\bullet)$ is the utility function of the subject. Probability distributions may then themselves be preferentially ordered by relative value, and decision making under uncertainty becomes a process of maximizing expected value over a set of possible prospects.

In a series of remarkably simple binary choice experiments between gambles of differing monetary value and probability of success, both the standard expected utility model (Allais 1953, 1979; Battalio, Kagel, and MacDonald 1985; Kahneman and Tversky 1979, 1984; Machina 1982, 1987; Schoemaker 1982; MacCrimmon and Larson 1979; Tversky and Kahneman 1981) and the more fundamental assumption of preference transitivity (Lindman and Lyons 1978; related results in Grether and Plott 1979; Pommerehne, Schneider, and Zweifel 1982; Ranyard 1976; Reilly 1982; Slovic and Lichtenstein 1983; Tversky 1969) have been called into question. People often prefer certain gains over gains with greater expected value that entail the possibility of acquiring nothing—the *certainty effect* (Kahneman and Tversky 1979, 1984). Yet they also prefer gambles with greater expected loss over certain losses of lesser magnitude when the former include the possibility of no loss—reflection (Kahneman and Tversky 1979, 1984). Of even greater import for a theory of preference, binary choices among gambles may be constructed such that intransitive cycles of the form $A > B > C > A$ are revealed (Lindman and Lyons 1978; Ranyard 1976; Tversky 1969).

We present a game-theoretic model of binary choice producing the above paradoxes by considering prospect choice not as a game against nature but as a veiled *n*-person game against opponents exposed to the same choice set as self, with success relativized to this set of presumed opponents. Prospect choice is then amendable to the standard Nash solution. The choice set defines the pure strategies available. Altering this set (i.e., introducing, eliminating, or substituting prospects) alters the game; because Nash solutions are contingent on strategic options, preference is contextualized to choice set.

There are several consequences. First, individual preference is not uniquely determined by choice set; rather, preference changes as the number of presumed opponents changes. Individual preference should then be exper-

imentally malleable to perceived competitive context (number of individuals presumed in the same choice situation), holding choice set constant. Second, the interface between decision theory and noncooperative game theory becomes reciprocal. Because preference is contingent on choice set and competitive context, prospect value is contingent on game context; yet once this context is specified, preferences are well-defined and should be amendable to experimental verification. Third, for some binary choice sets, a randomized equilibrium provides the only symmetric Nash solution to the n-person competitive game; thus variable behavior per individual need not imply an irrational actor or unstable preferences but simply a randomized strategy not measurable through current experimental design.

We also show that the game-theoretic assumption of *intelligence* (sensu Myerson 1991, 4; that is, players reason as game theorists under common knowledge) may be abandoned by embedding our logic in a simple evolutionary model. Besides subsuming the standard expected utility paradoxes, the evolutionary extension yields another insight: For some choice sets, the propensity to take sure bets over risky prospects of greater expected value may be enhanced by clustering competition among individuals playing the same (evolutionary) strategy. When a randomized strategy is the only symmetric Nash solution, preference is still subject to some degree of social control, even when the n-person game is fixed (i.e., both choice set and number of opponents remain unchanged), by structuring the nature of competition in evolutionary context.

2. GAMES AGAINST NATURE AS VEILED MALTHUSIAN COMPETITION

> Men are naturally envious, rejoicing in the shortcomings of their equals, and feeling pain at their virtues. For whenever a man conceives his own actions, he is affected with pleasure . . . in proportion as he can distinguish them from [the actions of] others, and regard them as something special. Therefore, a man will take most pleasure in contemplating himself when he contemplates some quality which he denies to others. (Spinoza [1673] 1957, prop. 55, pt. 3)

Suppose that individuals must play a game against nature in the form of some lottery chosen from a given set of alternative prospects. Once chosen, the prospect yields an associated outcome that affects the standing of the individual relative to others who have also had to choose from the same prospect set. For example, a prospect may list a business strategy's possible outcomes and associated probabilities. A realized outcome (e.g., a lucrative contract) influences the strategy holder's standing in her firm but

only after her colleagues' realized outcomes are also known. Relative success then translates into competition within the firm for, say, salary and/or promotion.

Although individuals are not involved directly in a common game, in that ostensible absolute payoff is not influenced by "opponent" play, the ultimate value of an outcome is contingent on prospect choice among future competitors. Each player maximizes her relative success knowing that rivals are exposed to an identical choice situation; prospects are ostensibly against nature, yet they influence the ultimate success of both self and opponent(s). Individuals are in intense *Malthusian competition*, where resources ultimately identified with success are inherently zero-sum, even though absolute competitive potential (ostensible prospect outcome) is not associated mechanistically with the decisions of rivals. We call this set of rivals plus self the *Malthusian arena* of the decision maker.

For simplicity, we consider an extended example of Malthusian arenas composed of three individuals. Our essential conclusions generalize to Malthusian arenas of any finite size (see Section 4). Individuals in our arena choose one of two prospects, R or S. Associate a nonnegative real number representing absolute competitive ability with each outcome of each prospect. These numbers might indicate monetary value (e.g., business contracts or grant awards) or some absolute measure of prestige (e.g., frequency of citation in academic journals or invitation to social gatherings). Once outcomes are realized, our three arena mates undergo common competition for access to a nondivisible resource that may be possessed by only one of them (e.g., a promotional slot).

We model two forms of Malthusian competition and compare their qualitative fit to experiment. First, Malthusian success is assumed to be a random draw among arena mates weighted by their relative competitive ability procured through realized prospect outcome. We call this *probabilistic competition* because a unique realized maximal outcome among arena mates does not insure competitive success for the prospect (strategy) associated with this outcome. This model induces the certainty effect and intransitive cycles but precludes reflection.

We then model Malthusian success as a random draw among those arena mates with maximal realized outcome. We call this *deterministic competition* because a unique maximal realized outcome among arena mates well defines the arena winner. This model induces the certainty effect, intransitive cycles, and one version of reflection (termed simple reflection herein; see Section 3), yet it precludes reflection through framing as described by Tversky and Kahneman (1981; Kahneman and Tversky 1984).

The latter form of reflection is precluded by the game-theoretic assumption of intelligence. Translating our model of deterministic competition into an evolutionary game (see Section 5) obviates the need for intelligent players while retaining the certainty effect, intransitive cycles, and simple reflection; in addition, reflection through framing can occur, producing all of the qualitative phenomena revealed in experiment.

Before considering each model in detail, we present conclusions common to both. Let π_j denote the probability that some individual playing prospect R will enjoy Malthusian success (deterministic or probabilistic) in an arena with j R and 3 – j S. By definition, $\pi_0 = 0$ and $\pi_3 = 1$. The probability that a specific R will enjoy Malthusian success in an arena with j R is then π_j/j (j > 0), 0 (j = 0). Denote by [j] the pure strategy 3-person outcome, where j individuals play R and 3-j play S. Because the game matrix is symmetric (i.e., outcomes are not unique to specific player choice), [j] is an equivalence class of outcomes up to relabeling of players. Then [j] is a Nash equilibrium (up to relabeling) in pure strategies iff (a) $(1 - \pi_j)/(3 - j) \geq \pi_{j+1}/(j + 1)$, j = 0, 1, 2, so no S does better by switching unilaterally to R, and (b) $\pi_j/j \geq (1 - \pi_{j-1})/(4 - j)$, j = 1, 2, 3, so no R does better by switching unilaterally to S.

We first consider when such pure strategy equilibria are *strictly preferred* by all players, so that any unilateral deviation leaves a player worse off than Nash play. All S is then a symmetric (strictly preferred) Nash solution in pure strategies ([0] is Nash) when $\pi_1 < 1/3$; as is all R ([3] is Nash) when $\pi_2 > 2/3$. These conditions are sufficient to generate the certainty effect and intransitive cycles under either deterministic or probabilistic Malthusian competition. Our analysis will also show that a strictly preferred Nash equilibrium in pure strategies is unique (up to relabeling of players) under both forms of competition.

2.1. PROBABILISTIC COMPETITION

2.1.1. The Certainty Effect

We begin by demonstrating the certainty effect. Let R have ostensible payoff u + w with probability p and w with probability 1 – p, while S yields the "sure bet" w + v with probability 1; hereafter we express these gambles as R = (u + w, p; w, 1 – p) and S = (w + v, 1), following Kahneman and Tversky (1979). The *baseline payoff* w ≥ 0 represents the absolute competitive ability enjoyed by each player prior to gamble choice. As we shall see, this prior competitive ability is nontrivial under probabilistic competition, as it influences the range of prospect pairs exhibiting the certainty effect

(Section 2.1.2). We assume w invariant across players, so differential competitive ability is solely a function of realized gamble. Suppose $pu > v > 0$, so R has greater expected payoff than S. Then, because gamble outcomes are realized independently among arena mates,

$$\begin{aligned}\pi_1 &= p[(w + u)/(3w + 2v + u)] + (1 - p)[w/(3w + 2v)];\\ \pi_2 &= p^2[(2w + 2u)/(3w + 2u + v)] + 2p(1 - p)[(2w + u)/(3w + u + v)]\\ &\quad + (1 - p)^2[2w/(3w + v)].\end{aligned} \tag{1}$$

We first focus on the case $w = 0$, $v = 1$, where prospect outcome completely determines competitive ability. The strategy class [0] (all S) is Nash when $\pi_1 < 1/3$ or, substituting in (1)

$$p < (u + 2)/(3u). \tag{2}$$

Because $(u + 2)/3u > 1/3$ for $u > 0$, [0] is a (strictly preferred) Nash equilibrium whenever $p < 1/3$, no matter how large u; choosing by maximal expected value, however, yields $p > 1/u$ as necessary and sufficient for preferring R. Even if gamble payoff increases without bound, the probability of winning the gamble must nevertheless be bounded away from zero if the gamble is to be preferred regardless of opponent play.

When $\pi_1 < 1/3$, all S ([0]) is the only pure strategy Nash solution (up to relabeling of players). For, if not, either [2] or [3] must be Nash; both possibilities entail $\pi_2 > 2/3$ when $\pi_1 < 1/3$ by condition (b) above. This is impossible. Suppose $\pi_1 < 1/3$, so $p < (u + 2)/(3u)$, $pu < (u + 2)/3$. For $w = 0$, $v = 1$, (1) gives $\pi_2 = p^2[2u/(2u + 1)] + 2p(1 - p)[u/(u + 1)] = 2pu(2u - pu + 1)/[(2u + 1)(u + 1)]$. Substituting $(u + 2)/3$ for pu, $\pi_2 < (2/9)(5u + 1)(u + 2)/[(2u + 1)(u + 1)]$. Because $u > 1$, this latter expression decreases with increasing u; π_2's least upper bound thus occurs at $u = 1$, giving $\pi_2 < 2/3$. Thus when [0] is a pure strategy Nash it is the only pure strategy Nash; a similar argument shows [3] (all R) to be the unique pure strategy Nash for $\pi_2 > 2/3$ (i.e., $\pi_2 > 2/3$ precludes $\pi_1 < 1/3$). When p is less than the bound given by (2) [1/3], sure bets are always preferred over gambles of higher expected value.

2.1.2. Preclusion of the Certainty Effect Under High Baseline Competitive Ability w

The general form of (2) (for $w, v \geq 0$, $pu > v$) precluding preference for R is

$$p < (v/3u)(3w + 2v + u)/(w + v), \tag{3}$$

which is equivalent to $\pi_1 < 1/3$ in (1). Denote by p_t that value of p making (3) an equality. As baseline payoff w becomes large for fixed u, v, $p_t \rightarrow v/u < p$ because $pu > v$. Although the certainty effect under probabilistic Malthusian competition exists for all finite w (i.e., $p_t > v/u$), its range vanishes as w is made arbitrarily large. When prospect outcome has negligible impact on baseline competitive potential (e.g., $w >> u$) individuals should not exhibit the certainty effect. This observation finds support in experiment; binary choices involving monetarily small outcomes [e.g., ($5, 1) vs. ($500, .01; $0, .99)] do not conform to the certainty effect (Kahneman and Tversky 1979), conforming, rather, to the logic of expected value. As we shall see, such sensitivity to baseline competitive ability is not possible under deterministic competition (see Section 2.2.2).

2.1.3. Asymmetric Equilibria in Pure Strategies

When $\pi_1 > 1/3$, $\pi_2 < 2/3$, no symmetric pure strategy equilibrium exists (see Section 2.1.1). The strategy class [2] is then a strictly preferred asymmetric equilibrium restricted to pure strategies iff $\pi_1 + \pi_2 > 1$ [by condition (b)]; similarly for [1] iff $\pi_1 + \pi_2 < 1$ [by condition (a)]. Both are possible, for appropriate p, u, yet not simultaneously satisfiable. Strictly preferred pure strategy equilibria are thus unique up to relabeling of players. This also implies that a strictly preferred pure strategy equilibrium is a *strong* Nash equilibrium (sensu Myerson 1991, 153; Bernheim, Peleg, and Whinston 1987), precluding collusion among players in the establishment of preferences.

Multiple pure strategy equilibria (beyond relabeling of players) exist only at $\pi_1 = 1/3$ or $\pi_2 = 2/3$. Both [0] and [1] are pure strategy equilibria at $\pi_1 = 1/3$. Because $\pi_1 = 1/3$ implies $\pi_2 < 2/3$, hence $\pi_1 + \pi_2 < 1$ [choose p in (1) so that $\pi_1 = 1/3$; substituting this value in the formula for π_2 in (1) gives $\pi_2 < 2/3$, so $\pi_1 + \pi_2 < 1$], neither [2] nor [3] are Nash. Similarly, $\pi_2 = 2/3$ implies $\pi_1 > 1/3$, hence $\pi_1 + \pi_2 > 1$. Thus only [3] and [2] are pure strategy equilibria at $\pi_2 = 2/3$.

2.1.4. Randomized Equilibrium

Without a coordination mechanism, asymmetric pure strategy equilibria are of limited value. Although our decision makers are intelligent, nothing in the prospect choice set assigns one prospect over another when asymmetric equilibria in pure strategies are strictly preferred. The dilemma is resolved by adopting a symmetric randomized equilibrium.

Let each individual play R with probability $0 < \psi < 1$, S with probability $1 - \psi$; that is, each plays the randomized strategy $\psi[R] + (1 - \psi)[S]$ using the notation of Myerson (1991). Denote by E(R), E(S) the probability that a specific individual wins (enjoys Malthusian success in the arena) when her realized play of $\psi[R] + (1 - \psi)[S]$ is R, S, respectively. Because her opponents also play $\psi[R] + (1 - \psi)[S]$, $E(R) = [\psi^2/3 + \psi(1 - \psi)\pi_2 + (1 - \psi)^2\pi_1]$, and $E(S) = [(1 - \pi_2)\psi^2 + \psi(1 - \psi)(1 - \pi_1) + (1 - \psi)^2/3]$. She has no reason to unilaterally deviate from the shared randomized strategy only if $E(R) = E(S)$. Because R is strictly inferior to S iff $\pi_1 < 1/3$, and S is strictly inferior to R iff $\pi_2 > 2/3$, the range permitting randomized equilibrium $\psi[R] + (1 - \psi)[S]$ is restricted to that of asymmetric equilibria in pure strategies, $\pi_1 \geq 1/3$, $\pi_2 \leq 2/3$ (see Section 2.1.3). Within this range, at least one ψ, $0 < \psi < 1$, must induce $\psi[R] + (1 - \psi)[S]$ as a symmetric Nash solution among players because any finite strategy symmetric game without a symmetric pure strategy equilibrium has such an equilibrium (Nash 1950). Further, because strictly preferred pure strategy Nash equilibria of our game are unique (up to relabeling of players; see Section 2.1.3), there is at most one such ψ when the pure strategy equilibrium is strictly preferred; a unique equilibrium in pure strategies implies a unique randomized equilibrium.

Within the range $\pi_1 > 1/3$, $\pi_2 < 2/3$, choice under uncertainty induces probabilistic behavior (randomized decision) as the only symmetric solution; here, the only endogenously coordinated response to uncertainty is uncertainty. As π_2 approaches 2/3, the cost of playing R (rather than S) decreases in arenas where an opponent is also playing R, so ψ approaches 1 as π_2 approaches 2/3. Similarly, ψ approaches 0 as π_1 approaches 1/3.

We do not insist that the symmetric (randomized) solution is inherently preferable to the asymmetric solution in pure strategies when both are possible. The latter, however, cannot arise without an exogenous coordinating mechanism. In this sense, the symmetric solution is mechanistically simpler. "Preference" for this solution would not be a matter of equilibrium efficiency (Pareto or otherwise) but, rather, a consequence of the casual structure of the world through which coordination must operate.

2.1.5. Intransitive Cycles

In this section, three prospects, R, S, and T, are constructed such that pairwise choice yields R > S, S > T, and T > R under probabilistic competition. Let $R = (u, p; 0, 1 - p)$; $S = (1, 1)$; $T = (v, q; 0, 1 - q)$. We have S > T whenever $q < 1/3$ because both $\pi_1 < q < 1/3$ and $\pi_2 < q^2 + 2q(1 - q) = 1 - (1 - q)^2 < 5/9 < 2/3$. Also, given any $p > 1 - 1/\sqrt{3}$, R > S for sufficiently large u because π_1

tends to $p > 1 - 1/\sqrt{3} > 1/3$ as u tends to infinity, and π_2 tends to $1 - (1 - p)^2 > 2/3$ (for $p > 1 - 1/\sqrt{3} > 1/3$) as u tends to infinity.

We now provide conditions that ensure T > R. The probability that one T will win against two Rs is $\pi_1 = q[p^2v/(v + 2u) + 2p(1 - p)(v/u + v) + (1 - p)^2] + (1 - q)(1 - p)^2(1/3)$, where we assume that if all in the arena lose their gamble, Malthusian success is an unbiased draw among the three participants. Similarly, the formula for the probability that two Ts will win against one R is $\pi_2 = q^2[2pv/(2v + u) + (1 - p)] + 2q(1 - q)[pv/(v + u) + (1 - p)] + (1 - q)^2(1 - p)(2/3)$. Fix $q = 2/7$, $p = 3/7$. Then S > T, and we can choose u so that R > S because $3/7 > 1 - 1/\sqrt{3}$. To show T > R, note that as v tends to infinity π_1 tends to $q + (1 - q)(1 - p)^2/3 = 374/1{,}029 > 1/3$. Also, π_2 tends to $q^2 + 2q(1 - q) + 2(1 - q)^2(1 - p)/3 = 746/1{,}029 > 2/3$ as v tends to infinity. Thus, for large enough v, T > R.

These constructed gambles are qualitatively similar to those used in experiment (Lindman and Lyons 1978; Ranyard 1976; Tversky 1969). S > T because, for a sufficiently unlikely outcome, certainty is preferred to uncertainty, regardless of prospect ordering by expected value. R > S because, for sufficiently likely outcomes, prospect preference is ordered by expected value. And T > R because, for two prospects with similar probability distributions, ordering is by expected value for sufficiently great differential expected outcome.

Each binary choice induces a unique Malthusian arena, potentially incommensurate with those induced by other binary choice sets. Intransitivity arises when preferences in one competitive context are illegitimately transferred to another context. Experimental procedure (necessarily) creates such separate decisional universes through the presentation of binary alternatives. Rather than being indicative of a fundamental inconsistency among preferences, experimental demonstrations of intransitivity may be interpreted as sensitivity to the competitive options provided by the experimenter. In this view it is not surprising that utility functions accessed via a series of binary choice gambles vary systematically with starting gamble (Hershey, Kumreuther, and Schoemaker 1982), as competitive context is defined by this gamble.

2.2. DETERMINISTIC COMPETITION

Before discussing reflection we describe the certainty effect and intransitive cycles under deterministic competition. Because the importance of differential outcomes is enhanced under deterministic competition, the range of gamble combinations exhibiting these phenomena increases.

2.2.1. The Certainty Effect and Intransitive Cycles

Again consider the prospects R = (u + w, p; w, 1 – p), S = (w + v, 1) with pu > v. Because any degree of competitive superiority insures success under deterministic competition, an R exposed to two Ss wins when her gamble is successful, so $\pi_1 = p$. Similarly, an S exposed to two Rs is successful only if both Rs lose their gamble, so $\pi_2 = 1 - (1 - p)^2$ The pure strategy equilibria conditions for strictly preferred Nash play are then

1. [0] (all S) is Nash iff $\pi_1 < 1/3$ iff $p < 1/3$ (the certainty effect)
2. [3] (all R) is Nash iff $\pi_2 > 2/3$ iff $p > 1 - 1/\sqrt{3} > 1/3$ (choice by maximal expected value)
3. [1] is Nash iff $1/3 < p < (3 - \sqrt{5})/2$ (asymmetric pure strategy choice, or a randomized symmetric equilibrium where S is chosen more often than R)
4. [2] is Nash iff $(3 - \sqrt{5})/2 < p < 1 - 1/\sqrt{3}$ (asymmetric pure strategy choice, or a randomized symmetric equilibrium where R is chosen more often than S).

Because $(3 - \sqrt{5})/2 < 1 - 1/\sqrt{3}$, as with probabilistic competition, when a Nash equilibrium in pure strategies is strictly preferred, it is unique. And, as before, when the pure strategy equilibrium is both strictly preferred and asymmetric, a unique randomized symmetric equilibrium exists.

The intransitive cycle demonstrated in Section 2.1.5 follows as well. There, we chose successful outcomes for our gambles as large as we liked, thereby mimicking deterministic competition, forcing pairwise choice leading to intransitivity to be solely a function of prospect probabilities when both gambles in the triple comparison are of greater expected value than the sure bet.

2.2.2. Irrelevance of Baseline Competitive Ability

Only one consequence of probabilistic competition discussed so far fails to carry over to deterministic competition. Because any degree of competitive superiority guarantees Malthusian success under deterministic competition, competitive ability shared among arena mates prior to prospect realization cannot affect Malthusian success. Unlike probabilistic competition, the certainty effect is not contingent on sufficiently low baseline competitive ability (Section 2.1.2). Thus absence of the certainty effect for sure bets of trivial value [e.g., ($5, 1); Kahneman and Tversky 1979] might suggest probabilistic over deterministic competition.

Because probabilistic competition precludes reflection but deterministic competition does not (see Section 3) we would rather not discard analysis of

the latter and so suggest another possibility. We have assumed that prospect outcomes are presented in terms of competitive ability. Yet outcomes must actually be presented as goods (e.g., money, vitae length, citation frequency, contract procurement). Individuals use these goods when vying for competitive success in a common Malthusian arena. There must then be a mapping of the quantitative value of a good, however defined, onto competitive ability. This mapping need not be linear. Intuitively, small amounts of a good appear unlikely to affect competitive ability. The Kahneman and Tversky (1979) example [(\$5, 1) < (\$500, .01; \$0, .99)] may reflect this. Such ancillary assumptions are less than ideal. Yet experimental analyses always require them and, in the present case, probabilistic competition is not immune because baseline competitive ability is an empirical matter. As we shall now see, reflection provides an independent means of comparing the empirical scope of the two forms of Malthusian competition.

3. REFLECTION

Two experimental designs demonstrate reflection. In one design, individuals are asked to choose between (u, p; 0, 1 – p) and (1, 1) with up > 1; when the certainty effect is evident, individuals (usually a different set of subjects; see Hershey and Schoemaker 1980) are then asked to choose between (–u, p; 0, 1 – p) and (–1, 1) for identical u, p; the modal preference is (–u, p; 0, 1 – p) even though –up < –1. The results are doubly paradoxical. Modal preference in both cases is for a prospect of lower expected value; yet risk aversion cannot account for both because choice under loss appears risk prone. We call this experimental design *simple reflection.*

In this design, the expected values of the sure bets are of opposite sign, as are the expected values of the gambles. Another experimental design shows that reflection may be induced even when the final outcomes associated with two sure bets and two gambles are identical, respectively. Give each individual u before asking them to choose between (u, p; 0, 1 – p) and (1, 1) with up = 1; they keep the prior gift of u (distinct from the outcome u associated with the gamble, which happens to have the same value as the gift) regardless of prospect choice. Again, a modal preference of (1, 1) appears. Now give each individual 2u before asking them to choose between (–u, p; 0, 1 – p) and (–1, 1). Modal preference is for the risky prospect (–u, p; 0, 1 – p). Here, both the certainty effect and reflection persist even when the sure bet and gamble are identical across choice sets, once the prior gifts have been factored into the outcomes. We call this design *reflection through framing*

(after Tversky and Kahneman 1981; Kahneman and Tversky 1984). Both simple reflection and reflection through framing appear in Kahneman and Tversky (1979).

Simple reflection arises under deterministic competition (Section 3.2) but is impossible under probabilistic competition (Section 3.4). Reflection through framing is impossible under both forms of competition because of the game-theoretic assumption of intelligence (Section 3.3). A simple evolutionary extension of deterministic competition, however, induces both simple reflection and reflection through framing while preserving the certainty effect and intransitive cycles (Section 5.2). The evolutionary model accounts, qualitatively, for all the paradoxes.

3.1. COMPETITIVE ABILITY WHEN PROSPECTS INDUCE LOSS

Although prospect outcomes are measured as gain or loss in absolute competitive ability, a negative net competitive ability is meaningless; someone with no competitive ability can never win in Malthusian competition. Prospect outcome can be negative, but realized competitive ability must be nonnegative if prospect choice is to be meaningful. We model this restriction by assuming that each player has competitive ability $w > 0$ prior to prospect choice. When a choice set is presented, this baseline ability is implicitly added to all outcomes. This baseline is not part of the experimental design, just as the size of the Malthusian arena is not part of the design. Thus a prospect presented experimentally as $(-1, 1)$ is interpreted as $(w - 1, 1)$ by our intelligent players, whereas $(1, 1)$ is interpreted as $(w + 1, 1)$.

3.2. SIMPLE REFLECTION UNDER DETERMINISTIC COMPETITION

Consider the prospect choice set experimentally presented as $\{S = (1, 1), R = (u, p; 0, 1 - p)\}$, $u > 0$. Given baseline competitive ability w, our intelligent players translate this choice set into $\{S' = (w + 1, 1), R' = (w + u, p; w, 1 - p)\}$. The simple reflection of the experimental presentation is $\{S^- = (-1, 1), R^- = (-u, p; 0, 1 - p)\}$ and is translated into $\{S^{-\prime} = (w - 1, 1), R^{-\prime} = (w - u, p; w, 1 - p)\}$. For $S^{-\prime}$ to be meaningful, $w > 1$; note that $R^{-\prime}$ does not require $w > u$; $w \leq u$ simply translates R^- to $R^{-\prime} = (0, p; w, 1 - p)$.

Suppose $S' > R'$ under deterministic competition, with $pu > 1$. In our usual three-person Malthusian arena, this is true when $p < 1/3$ (see Section 2.2.1). The simple reflection of these prospects may indeed be ordered as $R^{-\prime} > S^{-\prime}$. Because $S^{-\prime} = (w - 1, 1)$, $R^{-\prime} = (w - u$ [or 0 if $u > w$], $p; w, 1 - p)$, an $R^{-\prime}$ experiencing outcome w (with probability $1 - p$) precludes Malthusian success for any $S^{-\prime}$ in her arena but precludes her own success against any

$S^{-\prime}$ when experiencing outcome $w - u$ (with probability p), for $pu > 1$ implies $u > 1$. The probabilities associated with $R^{-\prime}$ Malthusian success are then reversed, with the unique Nash solution all $R^{-\prime}$ when $1 - p > 1 - 1/\sqrt{3}$ (see Section 2.2.1) or $p < 1/\sqrt{3}$, which is compatible with $p < 1/3$ for sufficiently small p. The certainty effect induces simple reflection under deterministic competition when $p < 1/\sqrt{3}$. Here, differential preference in variance ("avoiding" versus "seeking" risk) is an epiphenomenon; both choices are actually derived from the certainty effect (see Section 2.2.1) where different probabilities are associated with maximal gamble outcome. In this context, the terms risk avoidance and risk seeking are misleading.

When $1/\sqrt{3} < p < 1/3$, the certainty effect is exhibited for experimentally presented gains but not necessarily for experimentally presented losses. Here, equilibria in pure strategies are asymmetric under losses, with the only symmetric equilibrium a randomized strategy. Behaviorally, all individuals exhibit "risk avoidance" under gains, yet vary in their propensity to reflect.

If individuals coordinate their Nash equilibrium by using a randomized strategy, simple reflection will appear both inter- and intraindividually inconsistent. Evidence for interindividual inconsistency exists: Although experimental populations given choice involving loss exhibit reflection relative to populations given choice involving gain (i.e., $S > R$, whereas $R^- > S^-$ overall; Kahneman and Tversky 1979), individuals within a single experimental population vary in their propensity to reflect (Hershey and Schoemaker 1980). Other cases of inter/intraindividual inconsistency may be produced by examining the other inequalities associated with pure strategy equilibria presented in Section 2.2.1.

3.3. PRECLUSION OF REFLECTION BY FRAMING AMONG INTELLIGENT PLAYERS

The foregoing logic is inapplicable to reflection by framing. Under this experimental design, different prior gifts are presented for choice under gains versus losses, equalizing final outcomes between the sure bets (gambles) under gain and loss. Because our decision maker is intelligent, she will factor any prior gift into outcomes or, equivalently, baseline competitive ability; the choice problem is then invariant to presentation. Unlike simple reflection, baseline competitive ability (w) is not equal across the two choice problems because the prior gifts are not identical across choice sets. Knowing this, she sees no difference between the two sure bets or gambles, respectively. We differ further discussion until Section 5.2 ("Evolutionary Extension"). Clearly, this conclusion is invariant to the form of Malthusian competition.

3.4. IMPOSSIBILITY OF SIMPLE REFLECTION UNDER PROBABILISTIC COMPETITION

Consider the experimental presentation of $S = (1, 1)$, $R = (u, p; 0, 1 - p)$ which are translated by our intelligent players into $S' = (w + 1, 1)$, $R' = (w + u, p; w, 1 - p)$. Suppose $S' > R'$ with $pu > 1$, so that the certainty effect is exhibited. The experimental presentation for losses is $S^- = (-1, 1)$, $R^- = (-u, p; 0, 1 - p)$ which is translated into $S^{-\prime} = (w - 1, 1)$, $R^{-\prime} = (w - u, p; w, 1 - p)$; for $S^{-\prime}$ to be meaningful, $w > 1$ (see Section 3.2). Simple reflection (i.e., $R^{-\prime} > S^{-\prime}$) requires that the π_I associated with the choice set $\{R^{-\prime}, S^{-\prime}\}$ be greater than 1/3 in three-person Malthusian arenas (see Section 2). We show that this is impossible under the assumption that $S' > R'$ is a consequence of the certainty effect.

Under probabilistic competition, the π_I associated with $\{R^{-\prime}, S^{-\prime}\}$ is $\pi_I = p(w - u)/(3w - 2 - u) + (1 - p)w/(3w - 2)$. Keeping in mind that $w > 1$ for meaningful $S^{-\prime}$, $\pi_I > 1/3$ yields $pu < (3w - 2 - u)/(3w - 3) < 1$ whenever $u > 1$, which is insured by $pu > 1$, the latter given by our assumption that $S > R$ is due to the certainty effect. But $\pi_I > 1/3$ under reflection requires $pu < 1$, contradicting our assumption of the certainty effect under the original prospects. Simple reflection under probabilistic competition is thus impossible.

3.5. RISK SEEKING UNDER DETERMINISTIC COMPETITION

Section 3.2 argued that simple reflection under deterministic competition is not true risk seeking but, rather, risk avoidance under the certainty effect among prospects redefined through the inclusion of baseline competitive ability, so that net competitive ability is nonnegative. True risk seeking, however, does exist under deterministic competition. Any gamble R is then part of any Nash equilibrium under the choice set $\{R = (u, p; 0, 1 - p), S = (1, 1)\}$ so long as $u > 1$ (so that a successful gamble wins over the sure bet) and $\pi_I = p > 1/3$ (assuming three-person Malthusian arenas; see Section 2.2.1). This bounds the expected value of R by $up > 1/3$, not $up > 1$.

Probabilistic competition does not permit such risk seeking. Translating the preceding choice set by including baseline competitive ability $w \geq 0$, any R preference requires

$$\pi_I = p[(w + u)/(3w + 2 + u)] + (1 - p)[w/(3w + 2)] > 1/3$$

[equation (1) of Section 2.1.1], which may be written as

$$pu/(3w + 2 + u) + pw[1/(3w + 2 + u) - 1/(3w + 2)] + w/(3w + 2) > 1/3$$

$$pu/(3w + 2 + u) - puw/[(3w + 2 + u)(3w + 2)] + w/(3w + 2) > 1/3$$

$$pu(w + 1)/[(3w + 2 + u)(3w + 2)] > 1/3 - w/(3w + 2)$$

or $pu > (3w + 2 + u)/(3w + 3)$. Because $w \geq 0$, the left side of the latter expression is greater than 1 whenever $u > 1$, which is given,[1] insuring $pu > 1$. Both simple reflection and true risk seeking are thus limited to deterministic competition.

4. *N*-PERSON ARENAS

Suppose that a Malthusian arena is composed of n individuals. Then strictly preferred equilibria in pure strategies exist when (a) $(1 - \pi_j)/(n - j) > \pi_{j+1}/(j + 1)$, $j = 0$ to $n - 1$, so no S unilaterally switches to R, and (b) $\pi_j/j > (1 - \pi_{j-1})/(n + 1 - j)$, $j = 1$ to n, so no R unilaterally switches to S. Let $R = (u, p; 0, 1 - p)$, whereas S again is a sure bet of 1. Unconditional preference for S ([0] is an equilibrium) exists when $\pi_1 < 1/n$; similarly, $\pi_{n-1} > (n - 1)/n$ is equivalent to unconditional preference for R ([n] is an equilibrium). When $\pi_1 > 1/n$, $\pi_{n-1} < (n - 1)/n$, a randomized equilibrium provides the symmetric solution.

Under probabilistic competition, (some) R will be chosen when $\pi_1 = pu/(n - 1 + u) > 1/n$ or $pu > (n - 1 + u)/n$. As n tends to infinity, $(n - 1 + u)/n$ decreases to 1, recovering the logic of expected utility ($pu > 1$). Expected utility theory is thus a special case of probabilistic competition within arbitrarily large Malthusian arenas.

Expected utility theory is never consistent with deterministic competition. Here (some) R will be chosen when $\pi_1 = p > 1/n$ if $u > 1$; as arena size n increases to infinity, the expected value of R may be made arbitrarily low.

Although such preference might appear absurd, it is not inconsistent with actual behavior. State-run lotteries are usually a losing proposition when the cost of a ticket is compared to its expected value in the lottery. Jackpots are a function of ticket sales, so a high jackpot implies a large number of ticket holders. If a potential ticket purchaser sees herself in deterministic competition with others faced with the same decision, a large Malthusian arena is induced, permitting preference for gambles of considerably lower expected value than the sure bet of not purchasing a ticket. In this view, the jackpot itself is of secondary importance; it indicates a large Malthusian arena. Lottery sales should be successful when potential buyers perceive a large Malthusian universe; hence lottery advertising refrains from depicting wealthier elements of society as winners.

5. EVOLUTIONARY EXTENSION

5.1. THE EVOLUTIONARY EQUIVALENT OF INTELLIGENCE

Here, we present deterministic competition in an evolutionary framework to account for reflection by framing (defined in Section 3) as an experimental by-product of simple reflection. Suppose that a standard infinite population is partitioned into mutually exclusive, randomly formed groups of size n during some phase of a species' life cycle (after Wilson 1980). Deterministic competition within groups occurs during this partition phase, with only a single individual surviving to reproduce, so that selection operates via deterministic competition relativized to group membership.[2] For example, suppose that individuals vie for exclusive possession of territories necessary for reproduction. Territories can support no more than one reproductive individual; hence possession requires maximal intragroup competitive ability. Each territory is a patch so isolated that each individual must commit all her competitive effort in contest over a single territory.[3]

After selection in each group (e.g., competition leading to unique possession of a territory), the offspring of group members (e.g., the offspring of successful territory holders) disperse randomly and repeat the cycle. Individuals exhibit one of two strategies affecting their competitive potential: strategy S guaranteeing some standard competitive ability, and strategy R providing competitive ability u with probability p, nothing otherwise. Each individual possesses baseline additive competitive ability or "baseline fitness" w apart from strategy-induced abilities. Strategy inheritance is haploid genetic (e.g., Maynard Smith 1982; Boyd and Lorberbaum 1987) or cultural (e.g., Boyd and Richerson 1985). This scenario is an evolutionary equivalent of our game-theoretic model under deterministic competition, where the assumption of intelligence has been replaced by the assumption of strategy inheritance.

Evolutionary success is characterized by evolutionary stability, the persistence of a strategy at some nonzero frequency across evolutionary time when exposed sequentially to any admissible, randomly mixing, mutant strategy (Maynard Smith 1982). A set of concurrent strategies meeting this condition is collectively evolutionarily stable. When this set is a singleton, the evolutionarily stable strategy is equivalent to a symmetric Nash equilibrium (Myerson 1991, 122). When the set has greater cardinality, the evolutionary stable mix of pure strategies (or the randomized strategy equivalent to this mix) is equivalent to a randomized equilibrium (Maynard Smith 1982; Myerson 1991). When the Nash equilibria are strictly preferred, the converse of the foregoing two statements also holds by definition of evolutionary

stability (Maynard Smith 1982). Our game-theoretic conclusions for deterministic competition (see Section 2.2) thus transfer, inter alia, to our evolutionary scenario when equilibria are strictly preferred. When the symmetric equilibrium [0] ([n]) is strictly preferred, S (R) is the only evolutionarily stable strategy; when a (strictly preferred) randomized equilibrium is the only symmetric solution, the evolutionary population is behaviorally dimorphic.

The behavioral dimorphism may take two forms: a single randomized strategy played by everyone in the population with varying behavioral outcome or a strategic dimorphism where the population is composed of individuals playing pure strategies R and S. The relative frequency of pure strategies in the latter is identical to probability of pure strategy play in the former (Maynard Smith 1982). The latter does, however, have one advantage over the former: A random decision routine is unnecessary. Without such a decision routine, game-theoretic intelligence does not uniquely predict behavior when optimal strategy choice is frequency dependent (i.e., when symmetric pure strategy equilibria do not exist). Thus the evolutionary analog to intelligence (randomized decision) is open to experimental disconfirmation.

5.2. REFLECTION THROUGH FRAMING IN THE ABSENCE OF INTELLIGENCE

We have already noted that reflection through framing is incompatible with intelligence (see Section 3.3). Discarding intelligence through the evolutionary analog of our game-theoretic model of deterministic competition removes this difficulty. An evolutionary model does not require subject calculation per experimentally presented gamble. Choice evolves under natural ecology, shaping propensities to take sure bets over gambles. All the experimenter can do is document such propensity and attempt to infer the natural ecology (naturally recurring prospect choice sets) that affects evolutionary success. Just as the evolutionary decision maker "assumes" that competitors will be exposed to the same choice set as self, she "assumes" that any gift received without effort will also be bestowed on her rivals.

Behavioral propensities, however, are triggered by subjects' decision effort. The prior gift, passively received, is treated as a component of baseline fitness rather than part of decision context. The gift simply places a lower bound on baseline competitive ability, w; under deterministic competition, this baseline is relevant only if less than absolute value of the worst outcome among losses. Experimental design inducing framing by reflection precludes this possibility (Kahneman and Tversky 1979, 1984; Tversky and Kahneman 1981). Thus, in the absence of rational calculation, a "reflection through

framing" experimental design should be treated by subjects as if it were a "simple reflection" design, as revealed in the experiment.

5.3. CLUSTERING EFFECTS

Our evolutionary extension provides two further predictions beyond the purview of pure game theory. If a randomized decision routine is impossible (i.e., each individual must play a single pure strategy), the evolutionary importance of π_i is influenced by population size. In small polymorphic populations (with, say, m arenas of size n) a morph experiences a positive bias toward encountering the other morph, as sampling is without replacement. This inflates the evolutionary importance of π_1 (π_{n-1}) over π_{n-1} (π_1) for R (S) relative to an infinite population, leading to a positive bias in the relative frequency of R; for under deterministic competition, the evolutionary success of R is enhanced when R is distributed across groups rather than clustered within groups because a single R winning its gamble precludes S's competitive success in that group.

Similarly, a bias toward S can be induced by clustering individuals with the same strategy. When same-parent offspring experience a positive bias in sharing (n-person) arenas (i.e., the population is *viscous*, sensu Hamilton 1964; Myerson, Pollock, and Swinkels 1991), π_1 increases in evolutionary importance for S but decreases for R—conversely for π_{n-1}. The average S experiences fewer Rs winning their gamble, whereas the average R winning its gamble experiences more other Rs also winning gambles, thereby wasting R's competitive effort in competition with itself. In consequence, polymorphism is skewed toward S, whether strategic (pure strategy morphs) or behavioral (a randomized equilibrium).

6. CONCLUSION: RATIONALITY AND SOCIAL STRUCTURE

To date, discussion of the expected utility paradoxes preserves ontological primacy for the individual. This commitment is reflected in technical discussion of the functional form of utility (e.g., Fishburn 1982; Kahneman and Tversky 1979, 1984; Loomes and Sugden 1982, 1983; Luce 1991; Luce and Narens 1985; Machina 1982, 1987; Narens 1976); once this form is understood, individuals will be seen to conform to "rational" calculation (i.e., optimize expected utility). *Each individual can be characterized as a utility function in a monadic sense; the existence of others is not, in principle, a necessary condition for the identification of individual preference.*

Our thesis abandons this ontological priority. Individual preference is undefined without reference to a Malthusian arena of others. Once such an arena is specified, a utility function can be constructed. The construction is not, however, invariant per individual but subject to vagaries in the individual's social world. To the extent that Malthusian competition is influenced by local social structure, individual preference is inherently social; hence *game-theoretic* rationality is inherently social. Game theory and decision science are equally foundational, reciprocally defining one another.

Our analysis directs attention to the emergence and persistence of competitive structures. At first blush, this appears to make rational choice theory subservient to theories of structural change. The evolutionary approach of Section 5 has been used elsewhere to argue that this need not follow: Strategies designed to alter the competitive context of specific individuals may also evolve (e.g., Pollock 1988, 1991, forthcoming), providing "individually rational" competitive advantages. What is individually rational for one might induce change in the competitive world of others, forcing the latter to alter their behavior in a manner conforming to social structural explanation. The traditional debate over the ontological status of individuals becomes a nonproblem.

7. PROPOSITIONAL SUMMARY

1. Prospect payoffs are relativized to choice set.
2. Choice is not, however, invariant per choice set. One may also alter choice by altering the perceived size of the Malthusian arena.
3. For some choice sets the only endogenously coordinated response to uncertainty is to play a randomized strategy. For such sets, behavioral choice per trial is not intraindividually consistent without exogenous coordination.
4. Under probabilistic competition, prospect choice is also influenced by the competitive value of outcomes relative to baseline competitive ability (i.e., competitive ability prior to prospect choice). As outcome value(s) diminish relative to the baseline, choice is more likely to conform to ordering by expected value.
5. Probabilistic competition induces both the certainty effect and intransitive preferences but precludes any type of reflection.
6. Deterministic competition induces the certainty effect, intransitive cycles, and simple reflection but cannot account for reflection by framing under the game-theoretic assumption of intelligence.
7. Deterministic competition also induces true risk seeking, where some gambles of lower expected value than a sure bet (the range contingent on arena size) are preferred when some gamble outcome exceeds the sure bet.
8. Under probabilistic competition, as the Malthusian arena becomes arbitrarily large, prospects approach ordering by expected value.

9. Under deterministic competition, as the Malthusian arena becomes arbitrarily large, gambles of arbitrarily small expected value are preferred over sure bets so long as one gamble outcome with positive probability is greater than the value of the sure bet. This conclusion is consistent with the mass appeal of state lotteries, where the expected value of a lottery ticket is usually much less than its price.
10. An evolutionary interpretation of our model under deterministic competition holds that individuals do not calculate choice per experimental presentation but, rather, exhibit a propensity toward the certainty effect or simple reflection based on natural ecology. Experimental design should be informed by natural ecology.
11. If so, reflection by framing is experimentally equivalent to simple reflection. Prior gifts are ignored in choice, subsumed as part of baseline competitive ability. Choice is the product of an underlying propensity toward the certainty effect and is restricted to the choice set as such.
12. In our evolutionary model, when randomized strategies are technologically infeasible, the certainty effect is less likely as population size decreases, holding arena size and choice set constant.
13. In our evolutionary model, the certainty effect becomes more likely under viscous dispersal, holding population size, arena size, and choice set constant.

NOTES

1. Clearly, $u < 1$ would be disadvantageous for R under both deterministic and probabilistic competition.

2. Intragroup deterministic competition is a form of *soft selection* (sensu Wade 1985), where the carrying capacity of the group (here, one "winner") is invariant to the strategy composition of a group. Because competition is restricted to finite-sized groups, the certainty effect (i.e., preference for sure bets) is equivalent to selection for strategies with reduced variance in absolute fitness, as in Verner (1965) and Gillespie (1974).

3. This last assumption is included for presentational simplicity and may be relaxed in populations with finite dispersal distance.

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